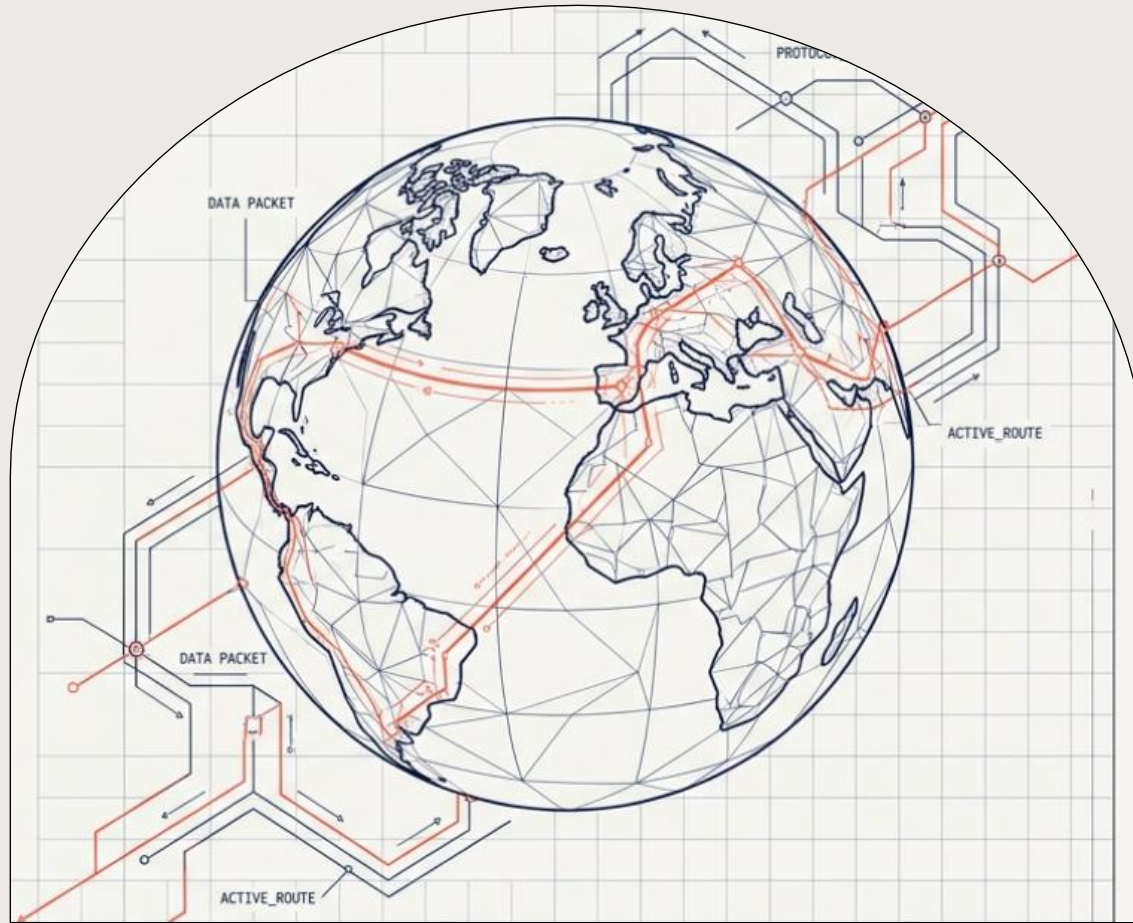


UNIT 1 COMPUTER SYSTEM



UNIT 1.1

Introduction to Computer System

COMPUTER

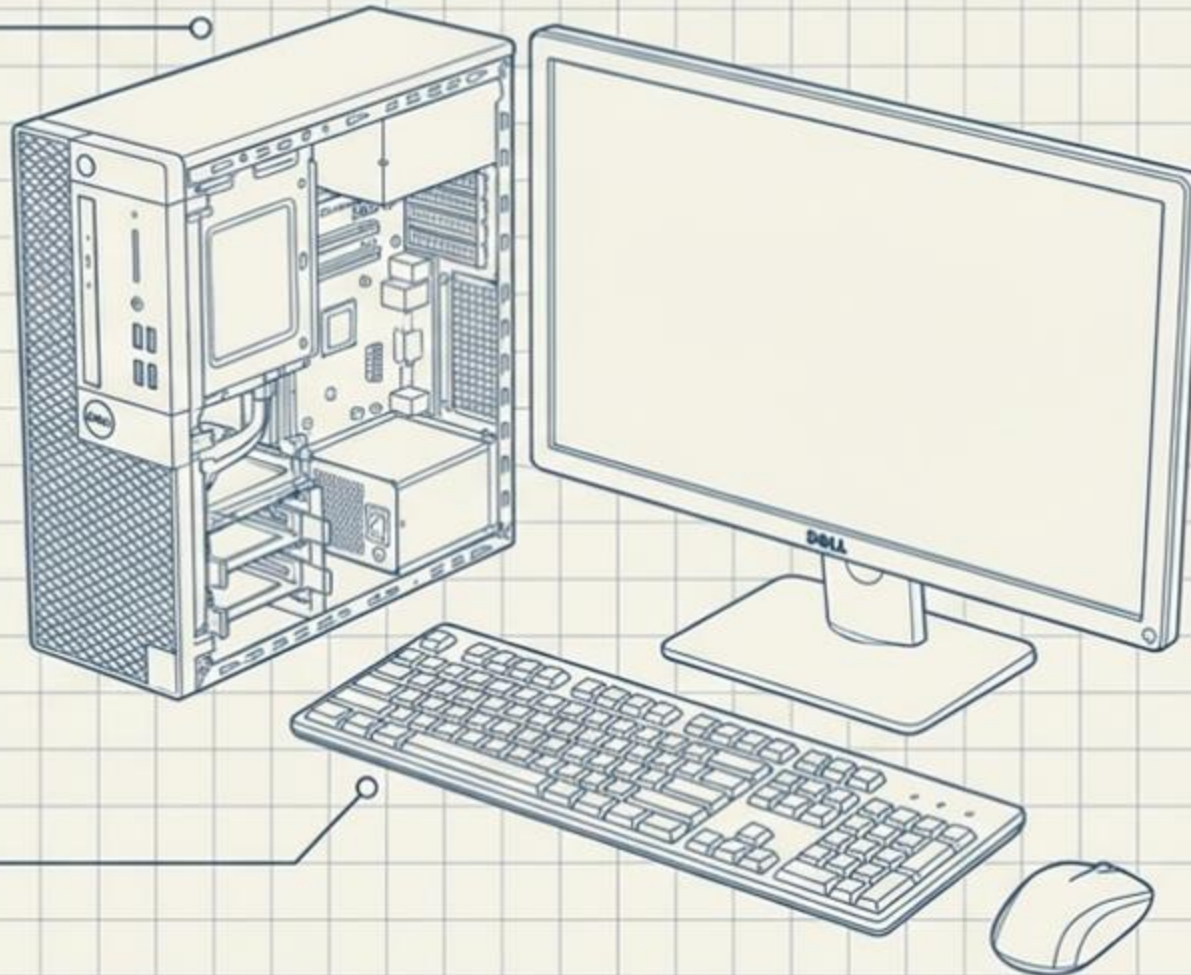
A computer is an electronic device that takes input from user, process it and produce meaningful output.

The Origin

**Computare*. Derived from Latin, meaning "to calculate."

The Impact

Once a tool purely for fast math, it is now the essential engine for documents, videos, games, music, and global data processing.

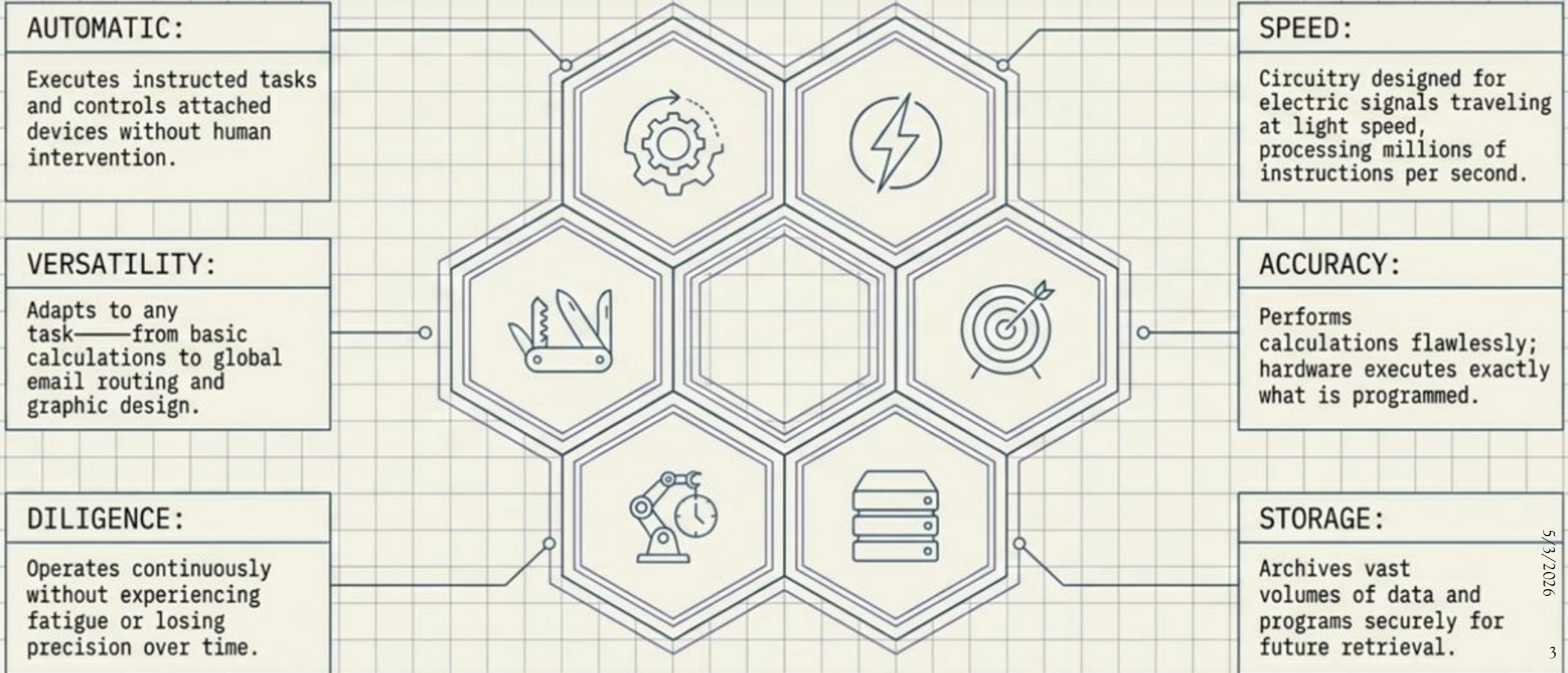


The Definition

A programmable electronic device that operates under stored instructions. It accepts raw input, processes it according to specific rules, and produces meaningful output at high speeds.

FEATURES OF COMPUTER

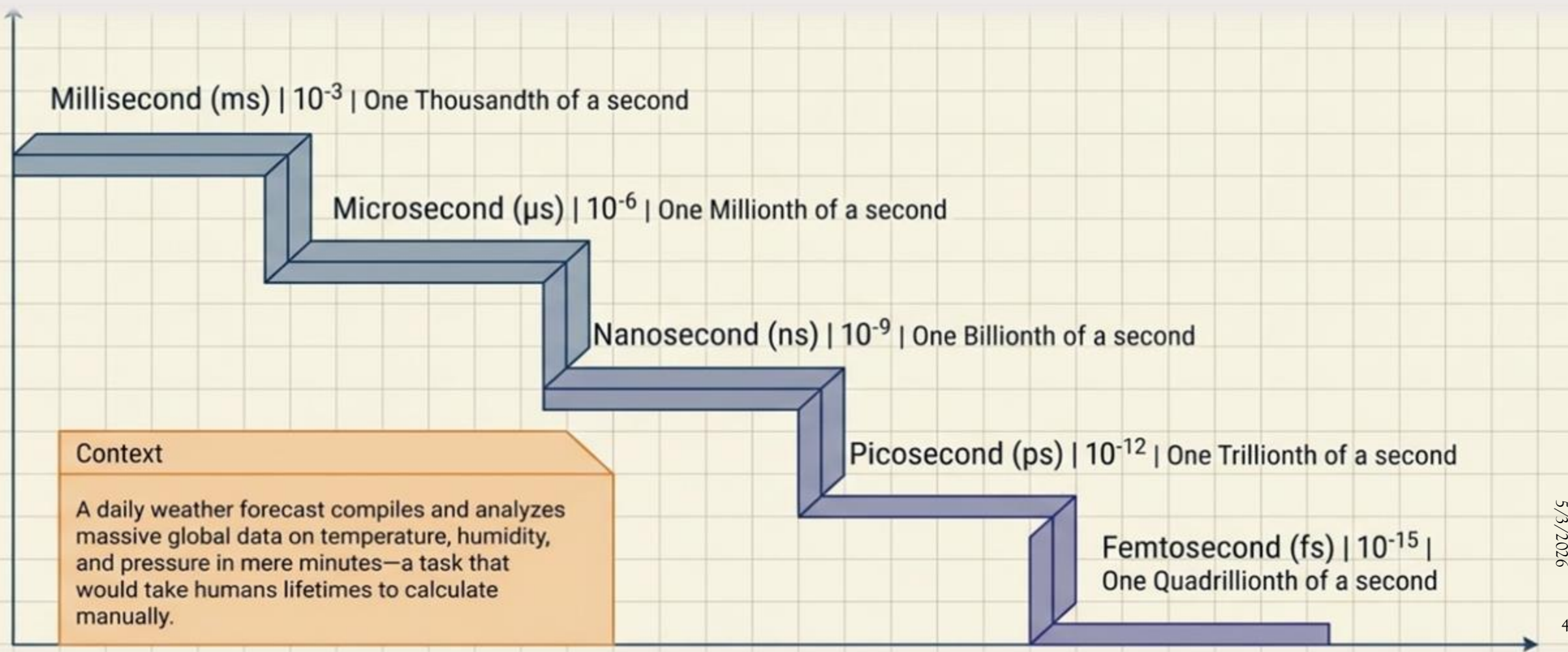
The Features of computer are as follows:



SPEED OF COMPUTER

The speed of computer is measured in seconds.

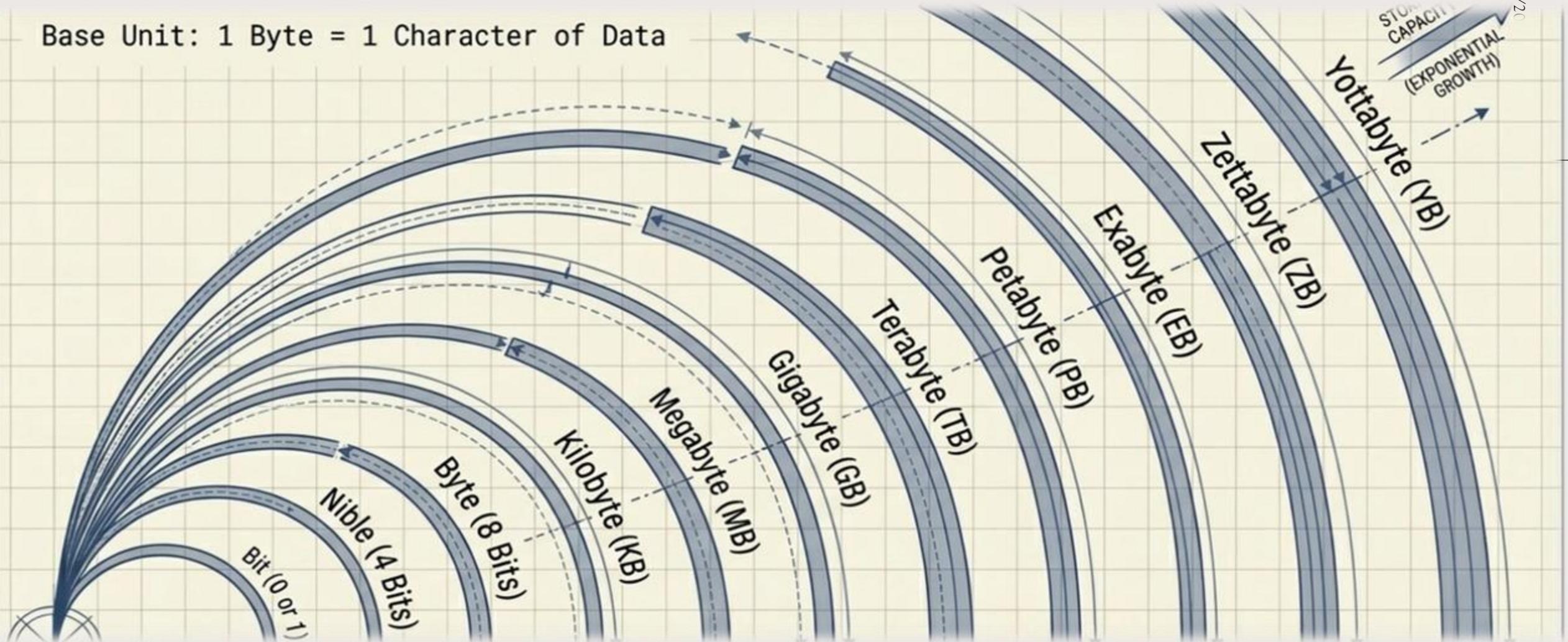
Unit of time	Part of the second	Power of 10
Millisecond	1/1000	10^{-3}
Microsecond	1/1000000	10^{-6}
Nanosecond	1/1000000000	10^{-9}
Picosecond	1/1000000000000	10^{-12}



DATA STORAGE

1 Bit = 0 or 1	8 Bit = 1 Byte	1024 Bytes = 1 Kilo Byte
1024 KB = 1 Mega Bytes	1024 MB = 1 Giga Bytes	1024 GB = 1 Tetra Bytes

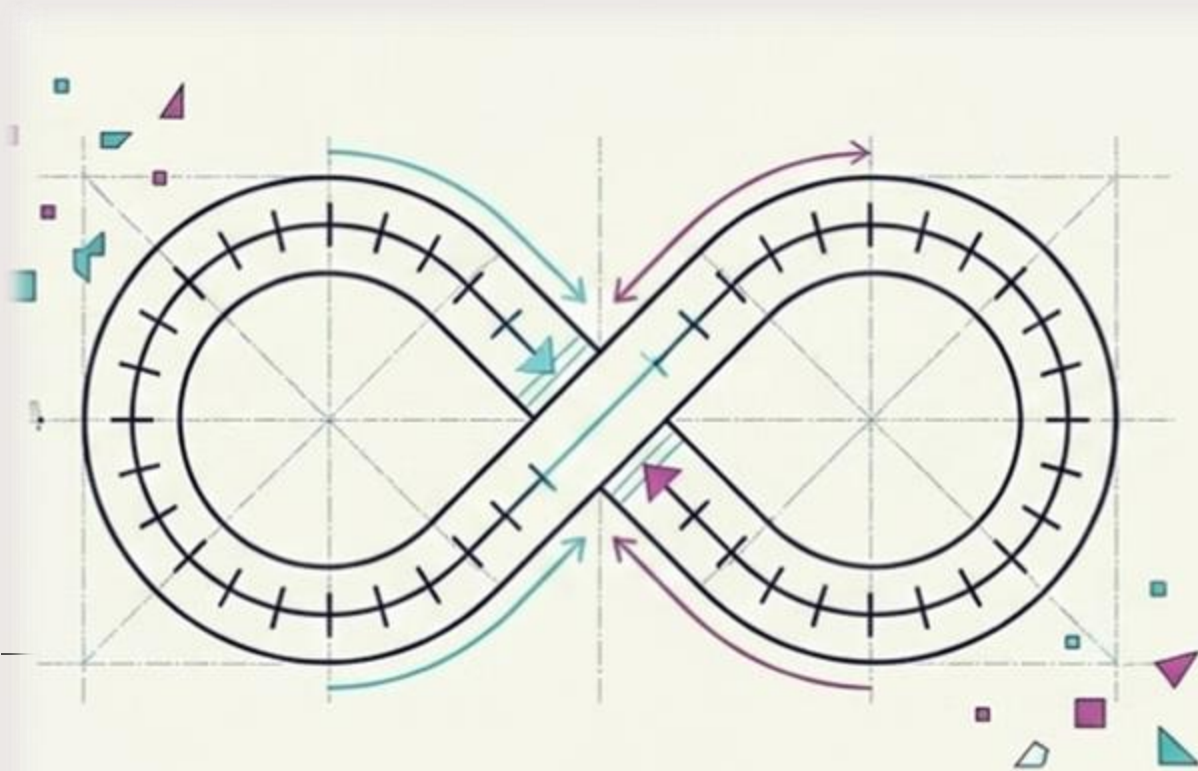
Base Unit: 1 Byte = 1 Character of Data



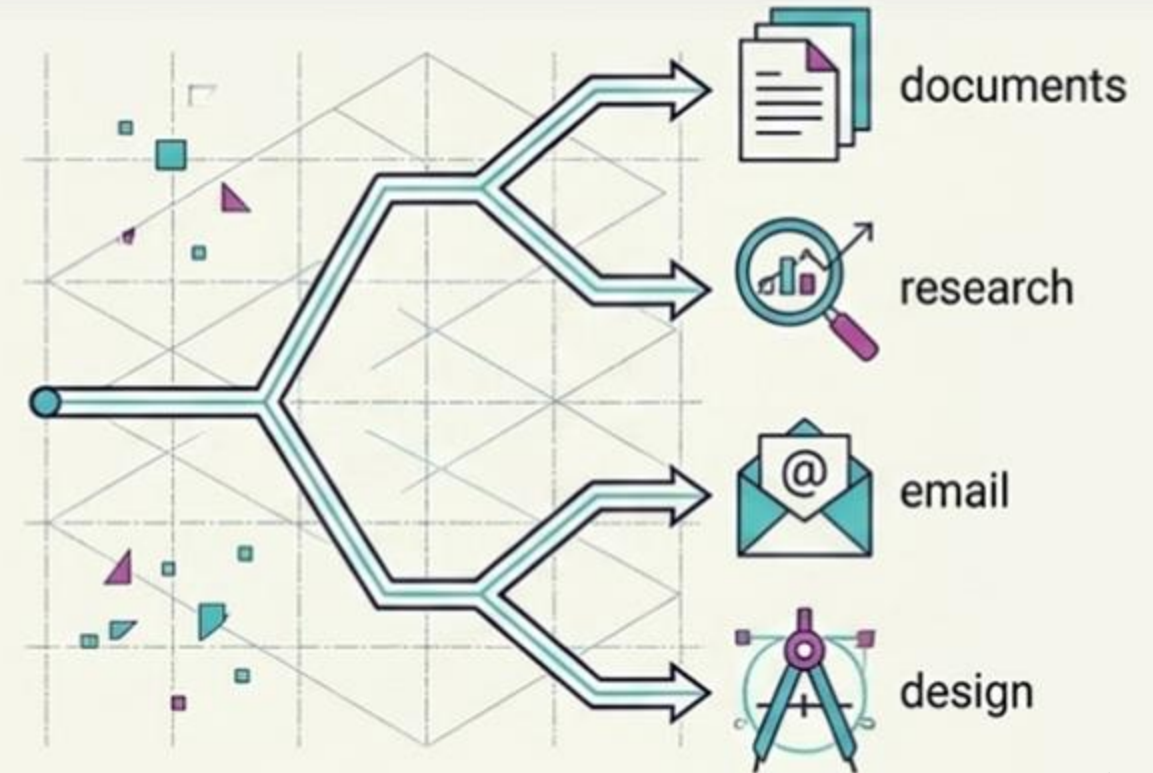
DILIGENCE & VERSATILE MACHINE

Diligence भनेको धेरै समय सम्म काम गर्दा पनि थकाई नलाग्ने ।

Versatile भनेको धेरै ठाउँहरूमा प्रयोग गर्न सकिने like in office, schools, bank, etc.



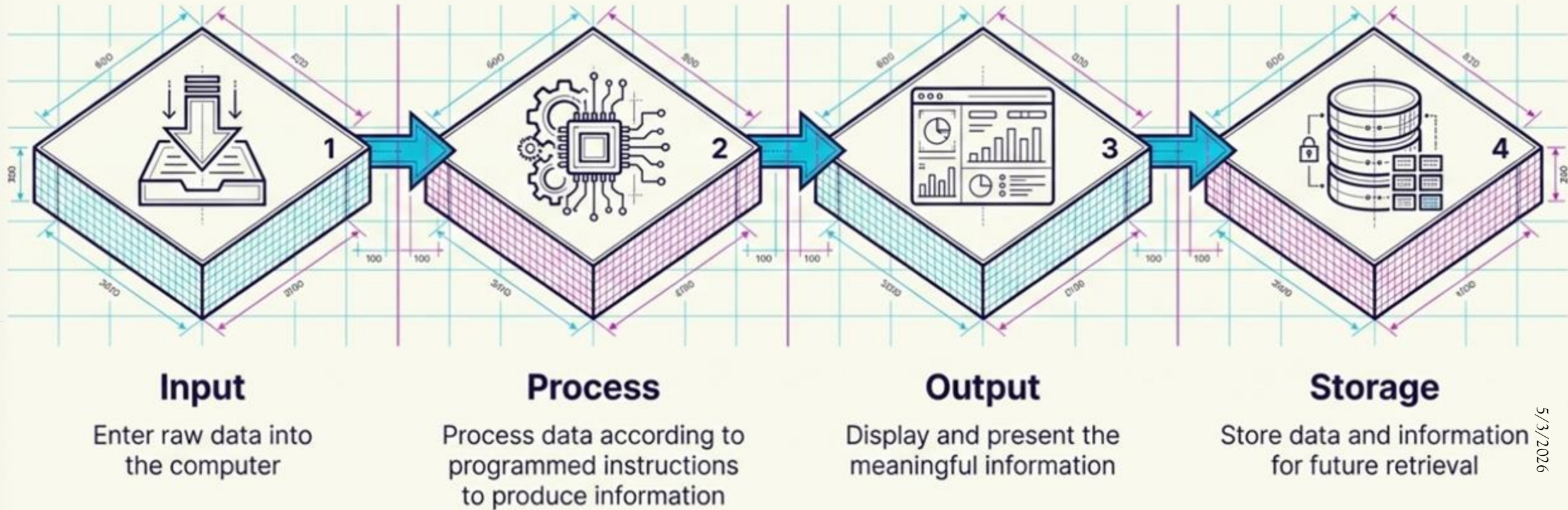
Diligence: The capability to perform tasks repeatedly over long periods without any degradation in speed or accuracy. Essential for quality control.



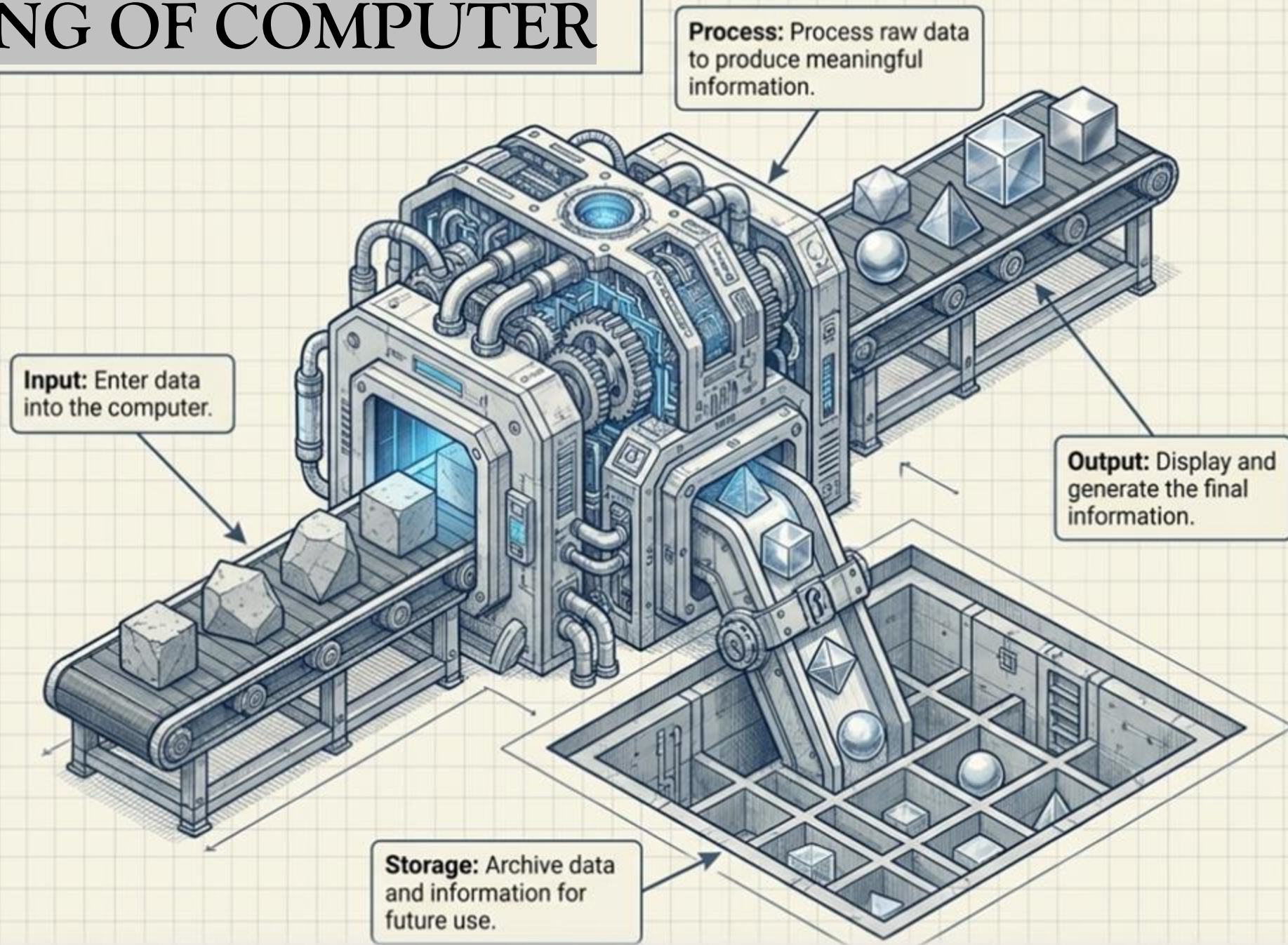
Versatility: The ability to execute entirely different kinds of tasks seamlessly—from sending faxes across the world to complex graphical design.

WORKING PRINCIPLE OF COMPUTER

The computer works on 4 different terms i.e. Input, Process, Output and Storage.



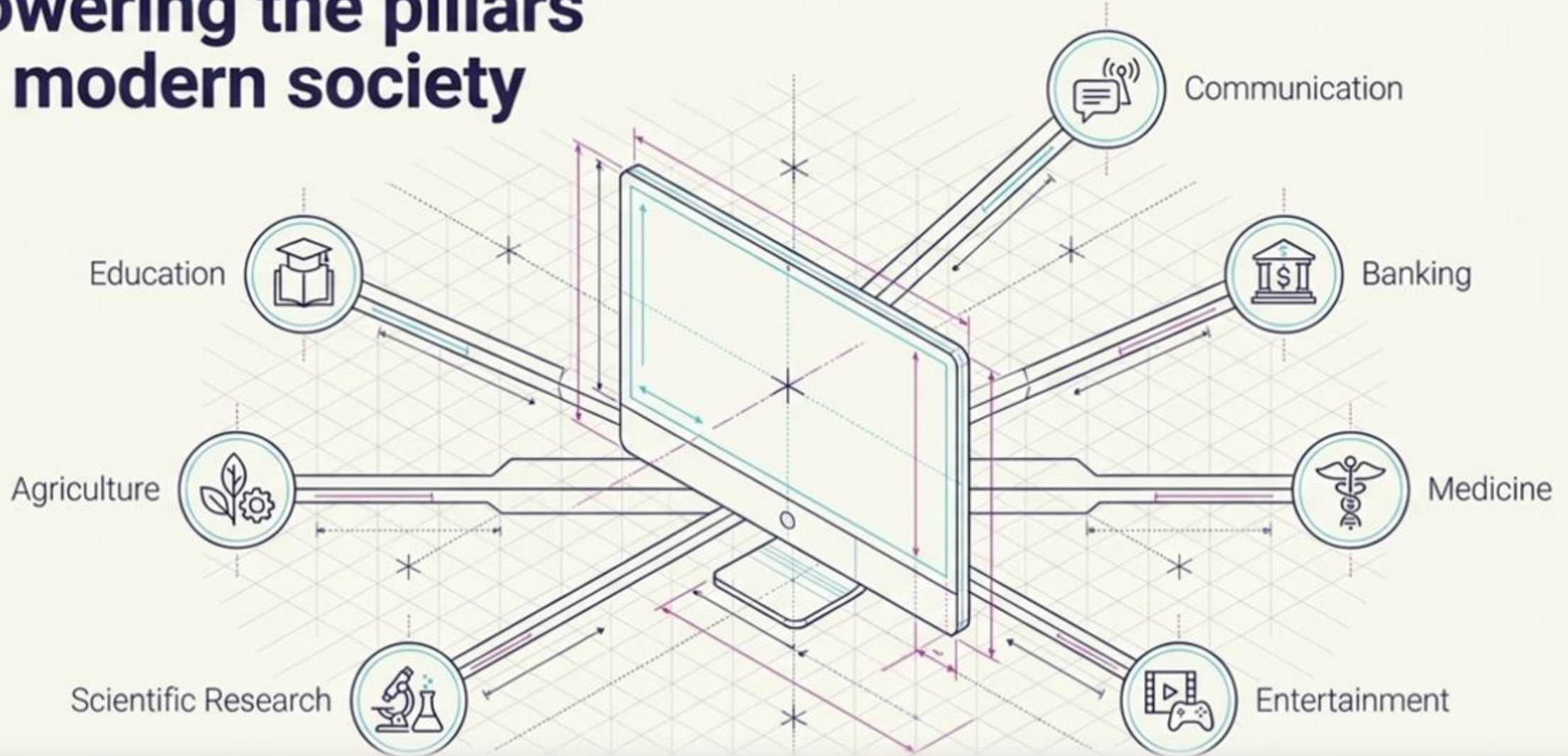
WORKING OF COMPUTER



APPLICATIONS OF COMPUTER

The applications of computer are as follows:

Powering the pillars of modern society



COMPUTERS USED IN EDUCATION

Below image shows how computer and technology has transformed traditional education method into modern education method:

- Unlocks internet-based learning and global subject research.

- Equips teachers to build highly effective multimedia presentations.

- Automates school administration, from student records to monthly billing statements and exam results.



COMPUTERS USED IN HOSPITAL

Below image shows how modern computing devices are used in hospital to check patients:



Diagnostics

Powers high-fidelity CT scans, ultrasounds, and ECGs for heart monitoring.

Management

Digitizes hospital patient records and medication tracking.

Research & Surgery

Analyzes massive datasets to develop new treatments, assists in active surgeries, and trains surgeons via advanced VR simulations.

COMPUTERS USED IN DIFFERENT SECTORS

The way how computer are used in agriculture, banking sector and entertainment is shown in the figure below:



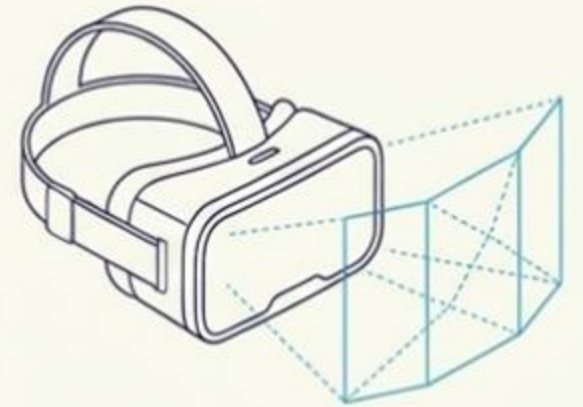
Agriculture

Revolutionizing traditional farming through precision agriculture, IoT sensors, drones, and data-driven decision-making to enhance crop yields.



Banking & Communication

Handling complex accounting, updating customer records, generating automated statements, and powering global social media and video conferencing.



Entertainment

Providing immersive experiences through powerful graphics, virtual reality, and vast libraries of streaming media.

IMPORTANT QUESTIONS

1. What is computer?
2. Mention the features of computer.
3. What is GIGO?
4. Why computer is called versatile machine?
5. Why computer is called diligence machine?
6. List any 4 applications of computer.
7. Write advantages and disadvantages of computer.

I) Speed of computer is measured in _____.

- a) Second b) Picosecond c) Nanosecond d) Millisecond

II) 1 GB is equal to _____ megabytes.

- a) 1024 MB b) 1048 MB c) 2048 MB d) 8 MB

III) Wrong output generated by the computer is called _____.

- a) Input b) Versatile c) GIGO d) Wrong data

IV) The smallest unit of data storage is called _____.

- a) Byte b) bit c) giga byte d) tetra byte

THE END